Technology Stack

Date	15 July 2026
Team ID	xxxxxx
Project Name	**Emotion Detection & Learning Support Engine**
Maximum Marks	2 Marks

Technology Stack

This document details the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs selected to implement the Emotion Detection & Learning Support Engine. The architecture balances local neural processing overhead with responsive visual tracking interfaces.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*Streamlit, HTML, CSS*	*Used to build an interactive and responsive web interface where students can enter learning problems, view detected emotions, confidence scores, mixed-emotion analysis, and analytics dashboards.*

2	**Backend / Server-Side**	*Python 3.12, Flask (support modules)*	*Handles emotion detection logic, model inference, API integration, session management, preprocessing, and communication between the UI and AI models. Python provides excellent support for machine learning libraries.*
3	**Database / Data Storage**	*SQLite, CSV Files*	*SQLite stores student emotion records, prediction history, and analytics data, while CSV files are used for training datasets and backup logging. Lightweight and suitable for local deployment.*
4	**Cloud / Hosting / Deployment**	*Google Gemini API, Hugging Face, Kaggle, Local Deployment*	*Gemini API generates personalized learning guidance, Hugging Face provides pretrained BERT models, Kaggle is used for model training, and the application runs locally through Streamlit.*

5	**Version Control & CI/CD**	*Git, GitHub*	*Enables version control, collaborative development, project backup, and source code management throughout the development lifecycle.*
6	**Third-Party APIs / Other Tools**	*Google Gemini API, PyTorch, Transformers, Plotly, Pandas, NumPy, Scikit-learn*	*Gemini API provides AI-generated learning guidance; PyTorch and Transformers power the BERT and BiLSTM models; Plotly creates interactive analytics dashboards; Pandas, NumPy, and Scikit-learn support data preprocessing, visualization, and model development.*